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IMPACT OF DELTA HEDGING FREQUENCY ON PERFORMANCE IN DIFFERENT VOLATILITY REGIMES

ABSTRACT

This project aims to uncover the effects of relaxing some of the assumptions made by delta hedging in simulated and practical settings to see if there's a strong contrast between theoretical assumptions and practical use cases. Moreover, it aims to uncover the influence of hedging frequency in such settings, recognizing the stability made by hedging and the profitability of such strategies. While there is no single metric to compare two hedging strategies in the general case, especially given variable tolerance for hedge error as opposed to total transaction costs, we use risk-adjusted metrics to draw reasonable conclusions. Ultimately, our findings show that optimal hedging is complex and highly contextual depending on market conditions and transaction costs, with adaptive hedging strategies considering volatility regimes generally outperforming static strategies.

INTRODUCTION

The goal of delta hedging is to stay neutral to price movements in the underlying. To implement this, a common strategy is to buy calls and sell shares in the ratio determined by the delta of the call option. However, some of the key assumptions made under delta hedging include, the trader can continuously hedge to balance the position, the option and underlying are both liquid, there is a constant volatility, and there are no transaction costs.

To break some of these assumptions, we utilize a GARCH(1,1) model to compare the performance of delta hedging in different volatility regimes. A GARCH(1,1) is used over a different parameterized model like GARCH(2, 1) for instance, because of some of the simplistic assumptions made by the simulation. GARCH(1, 1) uses a single lagged square returns, and a single lagged conditional variance, whereas GARCH(2, 1) uses 2 lagged square returns coefficients. This makes GARCH(2,1) a better model to simulate the volatility of an underlying by overfitting on past returns strongly, but it isn't great when the simulated data doesn't realistically have past returns, and overfitting on the autoregressive values would make the model return more stable volatility changes.

To evaluate the various delta hedging frequencies, we define risk-adjusted metrics that capture both the accuracy and efficiency of the hedge. First of all, the Sharpe ratio of the hedged

portfolio serves as a key measure of risk-adjusted returns, reflecting how effectively the hedge mitigates price fluctuations while maintaining a stable return profile.

$$\text{Sharpe Ratio} = \frac{E[R_p - R_f]}{\sigma_p}$$

R_p is the portfolio return, R_f is the risk-free rate, and σ_p is the standard deviation of returns. The Sharpe ratio is used to assess the risk-adjusted performance of a hedging strategy. Artur Sepp (2011) applied this metric in his analysis of delta-hedging strategies with discrete hedging and transaction costs. A higher Sharpe ratio typically suggests that a hedging strategy reduces risk more efficiently than it erodes returns. Additionally, we use the hedge mean squared error (MSE) to quantify the overall accuracy of the hedge. Hedge MSE measures the deviation between the hedged portfolio's value and the ideal perfectly hedged value, providing a direct assessment of hedge performance.

$$\text{Hedge MSE} = \frac{1}{N} \sum_{t=1}^N (\Delta P_t - H_t)^2$$

In this formula, ΔP_t is the change in the option price while H_t is the hedging adjustment at time t . By minimizing hedge MSE, we aim to identify the hedging frequency that most effectively neutralizes directional risk. These metrics offer a balanced evaluation framework: the Sharpe ratio accounts for the tradeoff between return stability and risk, while hedge MSE captures the precision of the hedge itself. Together, they allow us to assess the impact of volatility dynamics on delta hedging effectiveness systematically.

Furthermore, in a practical setting, transaction costs are crucial in determining the optimal hedging frequency. We see a fundamental tradeoff: hedging too infrequently exposes the portfolio to accumulating directional risk. However, hedging too frequently results in excessive transaction costs that may outweigh the benefits of reduced hedge error. Given that neither extreme is optimal, there should exist a Pareto-optimal tradeoff between transaction costs and hedging error, offering a clearer understanding of how transaction costs shape the effectiveness of delta hedging in realistic market conditions.

RELATED WORK

The work of Black and Scholes (1973) established the foundation of modern option pricing and delta hedging by assuming continuous rebalancing, no transaction costs, and constant volatility. Merton (1973) extended this framework to continuous-time trading, but both models failed to account for the practical limitations that traders face, such as stochastic volatility and market frictions. Recognizing these limitations, Hull and White (1987) explored how stochastic volatility affects option pricing, breaking the assumption of constant volatility. Their model was

one of the first to incorporate a volatility term that evolves over time, making it a precursor to later models that integrate more dynamic market conditions.

Building on these foundational models, researchers have improved delta hedging strategies by incorporating realistic market conditions, such as transaction costs and stochastic volatility. The Heston (1993) model introduced stochastic volatility, which allows for a more accurate representation of market behavior than the constant volatility assumption in Black-Scholes. Bakshi, Cao, and Chen (1997) provided empirical evidence that stochastic volatility models offer better hedging performance than traditional Black-Scholes-based approaches.

Taking dynamic volatility a step further, Engle (1982) first introduced the ARCH model, which was later generalized by Bollerslev (1986) into the GARCH framework. These models have been extensively used to capture volatility clustering, where periods of high volatility are followed by further high volatility. Christoffersen and Jacobs (2004) demonstrated that GARCH-based option pricing models improve hedging performance by aligning volatility estimates with observed market behavior. Since our methodology relies on a GARCH(1,1) model to simulate volatility regimes, these works directly inform our approach.

A critical challenge in delta hedging is balancing hedge accuracy with transaction costs. Leland (1985) was among the first to show that continuous delta hedging is infeasible when transaction costs are introduced. His work demonstrated that a modified volatility adjustment could approximate the effects of continuous rebalancing in the presence of proportional transaction costs. Boyle and Vorst (1992) extended this by deriving an optimal hedging frequency based on a power-law relationship between volatility and transaction costs. Their findings suggest that hedging should be more frequent in high-volatility environments and less frequent when transaction costs are high. By testing different hedging frequencies, we explore this principle in our simulation.

More recent research has emphasized adaptive hedging strategies, which adjust rebalancing frequency based on market conditions. Zakamouline (2006) developed a framework for dynamically adjusting hedge frequency, showing that adaptive strategies outperform static approaches under changing volatility conditions. Similarly, Fouque, Papanicolaou, and Sircar (2000) analyzed hedging strategies under different market regimes, highlighting the advantages of adjusting hedging frequency in response to shifts in implied and realized volatility. These insights directly align with our investigation of how different hedging frequencies perform under varying volatility conditions.

Furthermore, empirical studies have played a crucial role in validating theoretical hedging models. Guéant, Lehalle, and Fernandez-Tapia (2013) investigated optimal execution and hedging strategies, considering market impact and transaction costs. Their work highlights the importance of balancing trade execution with hedge accuracy, a concept we incorporate by evaluating P&L variance and transaction cost tradeoffs in our simulations.

Several studies have also used historical market data to test hedging strategies. Dumas, Fleming, and Whaley (1998) evaluated the effectiveness of hedging under real-world market conditions, finding that implied and realized volatility differences significantly impact hedge performance. Our empirical analysis follows a similar approach, using Russell 2000 index data.

Additionally, research into the probability distribution analysis of hedging errors has been conducted to assess risk exposure under different rebalancing strategies. Glasserman (2003) explored Monte Carlo methods for financial engineering, including their applications to risk management and option pricing. Our work extends this by simulating 100 hedge portfolios and estimating their probability density functions using histogram-based kernel density estimation. This allows us to empirically analyze the distribution of hedge errors and assess risk-adjusted performance under varying rebalancing strategies.

METHODS

Our main strategy to create the simulation involves setting up the returns and spot volatilities for the underlying asset over a period of 252 days, i.e. a single trading year. We then use those returns to simulate the price movement, calculate call prices over each discrete time step in that period, and calculate the relevant delta for hedging. We re hedge after a set number of time steps (1, 10, 50, 100), shorting shares to offset the compute delta at the relevant step. The simulated portfolio value is created by adding the price of the call options, subtracting the price of the shares we sold, and adding the cash that was gained or lost by rehedging at a different delta value.

To look at the relevance of our findings, we also produce probability distributions over one hundred simulated examples. The distributions are created by binning the results from each simulation into a histogram, based on the P&L values calculated by looking at the initial vs. final value of the portfolio. Once we have computed the counts for each of the bins, we use Gaussian kernel density estimation (KDE) to produce a smoothed pdf. By creating these PDFs, we expect to see more stability for higher hedging frequencies than for lower frequencies, especially for high volatility regimes. To test our hypothesis, we began by testing with constant volatility to get a baseline. In order to represent realistic volatilities, we choose $\sigma = 0.1$ to represent lower volatility (eg. broad market indices) and $\sigma = 0.5$ to represent a high volatility regime (eg. meme stocks and speculative assets). Following this, we use GARCH(1, 1) to generate stochastic volatility in different regimes.

We also compare the findings from a historical perspective by looking at the Russell 2000 during the periods of November 10, 2022 - January 31, 2023. We take the same expiry for all the calls on January 31, 2023, and an at-the-money strike of \$1850 for consistency. We also use a rolling window of 10 days of stock price, to compute spot realized volatility to understand our volatility regime, and compare it against the implied volatility of the calls. We use the delta implied by the calls to find our hedge.

RESULTS

As a control case, we first simulated constant volatility. We found that all the hedging strategy portfolios actually converge to about the same value. This is indicative of the fact that our initial hedge doesn't need to be reheded too strongly, and by expiry, the delta is at 1 or 0. If it's at 1, the value (basically equal to payoff) of the call is exactly the same as the payment for the shorted stock, indicating the 0 payoff. Similarly, at 0 delta, the payoff of the call is 0, and since the delta is 0, we don't need to hedge any shares. This would be an ideal case where implied and actual volatility are the same and are constant.

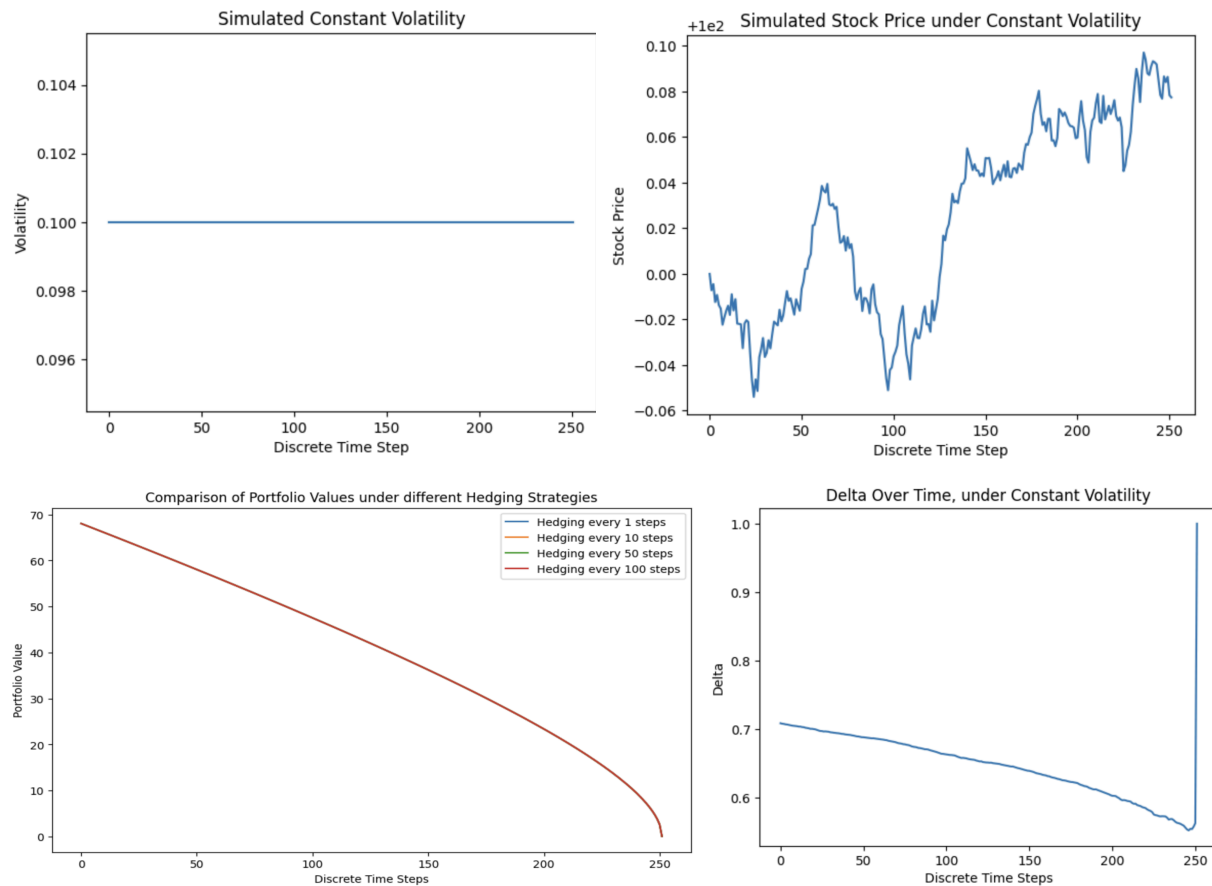


Fig 1: Simulation Under Constant Volatility. Portfolio Converges to 0, and the delta of call moves up to 1 (ITM call)

Moving onto the simulation of GARCH, we actually showcase 2 simulations, one where the call is in-the-money at expiry, and one where it is out-of-the-money. A key thing to note is that regardless of the scenario, the frequent hedging seems to converge a lot smoother than any of the other frequency hedging strategies, and the 100-steps hedging strategy is the most volatile to the constantly changing delta, that it isn't able to hedge against (the portfolio values at each of these points isn't just the same at the least common multiple of 1, 10, 50, and 100, because we count

cash into the equation). We can also recognize that less frequent hedging strategies, like every 10 steps, can actually be somewhat ideal since you can benefit from all the minor gamma exposure that you can't by frequent hedging where you realize minor profits and losses. This recognizes the expected tradeoff between profitability and stability in delta hedging frequency.

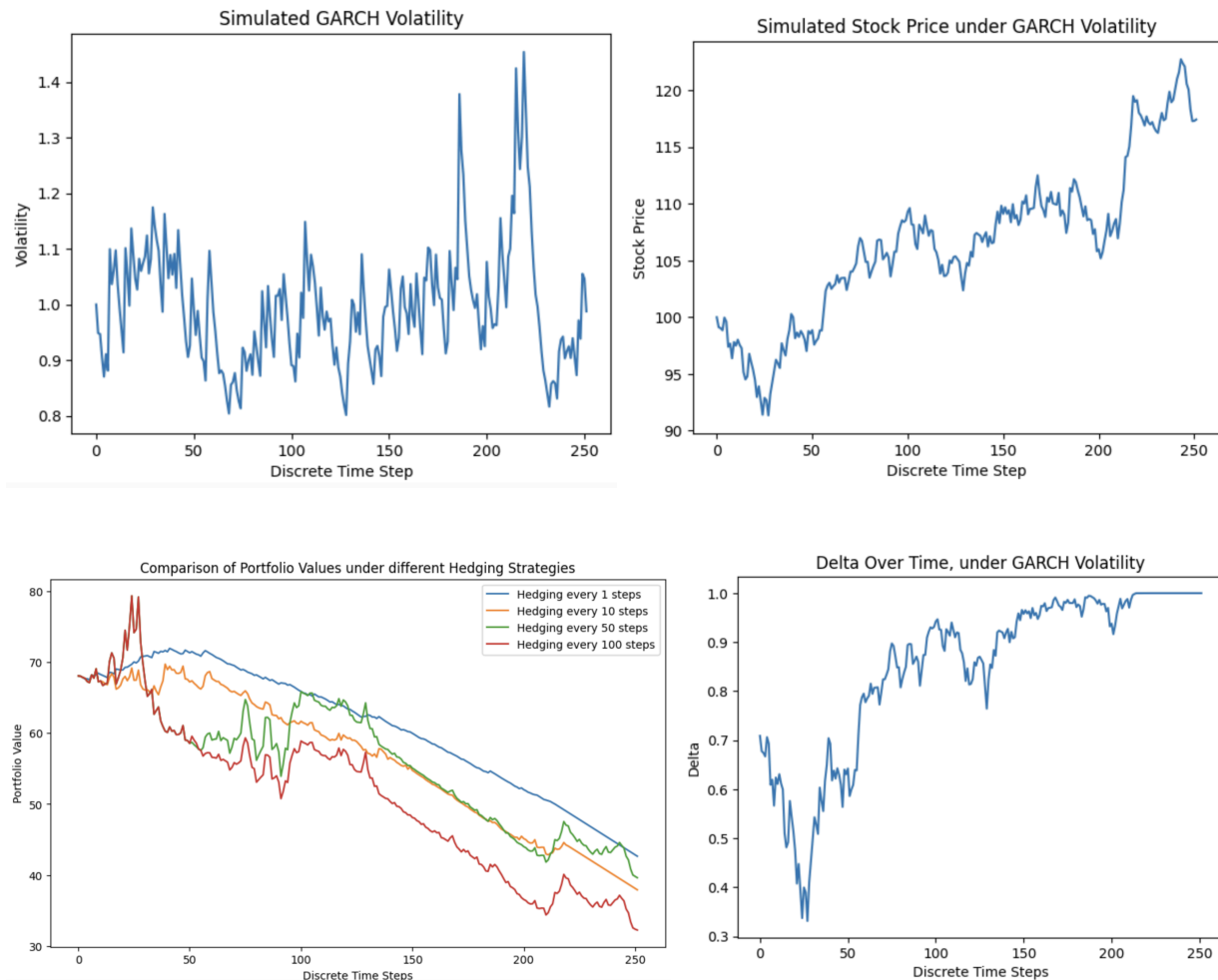


Fig 2: Simulation Under GARCH (1,1) Volatility (ITM Call). The portfolio value moves much smoother under every step hedging (blue line), and the other hedging frequencies are lot more volatile to the underlying asset movement and volatility

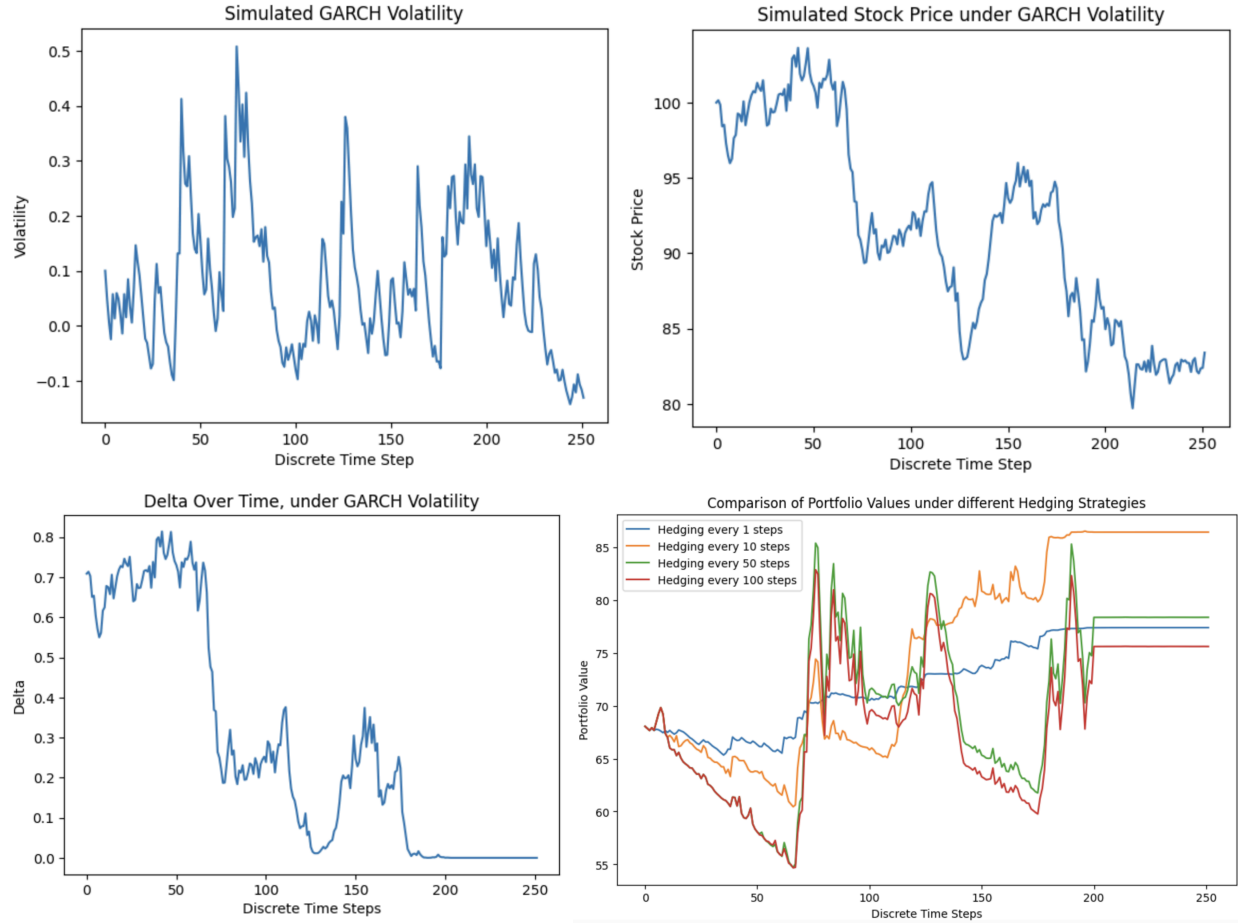


Fig 3: Simulation Under GARCH (1,1) Volatility (OTM Call). Delta ends at 0. The portfolio value moves very differently for all the portfolio values because of the gamma exposure.

After running individual simulations, we scaled up to repeating trials over multiple runs to get a general sense of the effects of hedging at different frequencies. Note that metrics describing the PDFs can be found in Appendix A. Looking at Figure 4 and Figure 5 for constant volatility, we find that the means are aligned under both volatility regimes, although the differences between the distributions are quite drastic under high volatility as opposed to low volatility. This suggests, as we expected, that hedging more frequently is more critical in higher volatility regimes as opposed to lower volatility regimes. However, looking at the case of $\sigma = 0.5$ in Figure 5, we note that the dropoff between a hedging frequency of 10 and 50 is much more drastic than hedging at a frequency of 1 vs 10, or even 50 vs. 100. In some ways, this suggests that hedging more frequently or less frequently has diminishing returns in either direction.

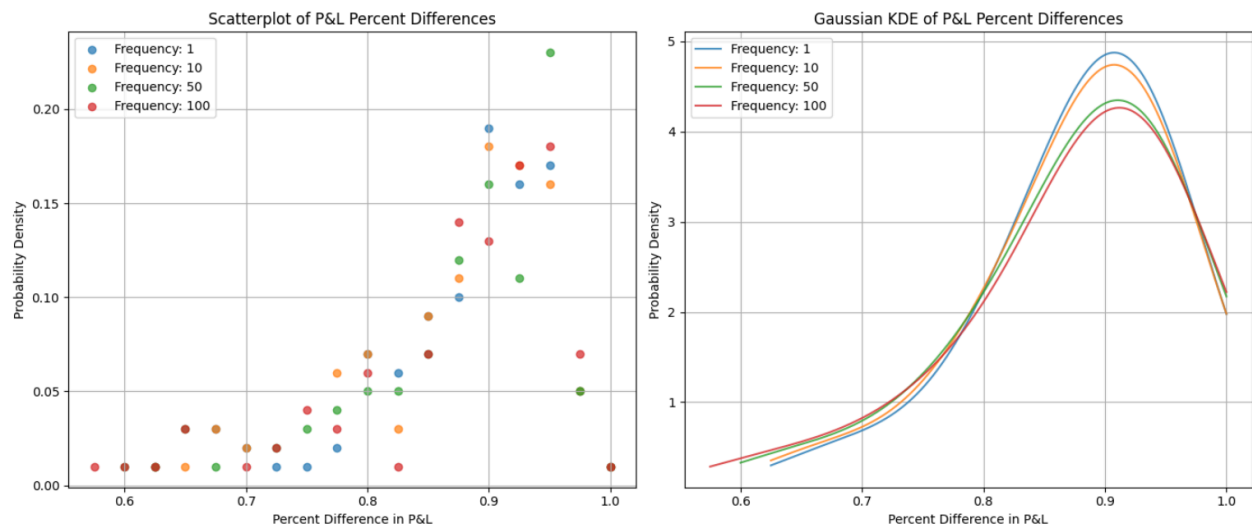


Fig 4: Distribution of percent differences in P&L comparing initial and final portfolio value for constant volatility $\sigma = 0.1$

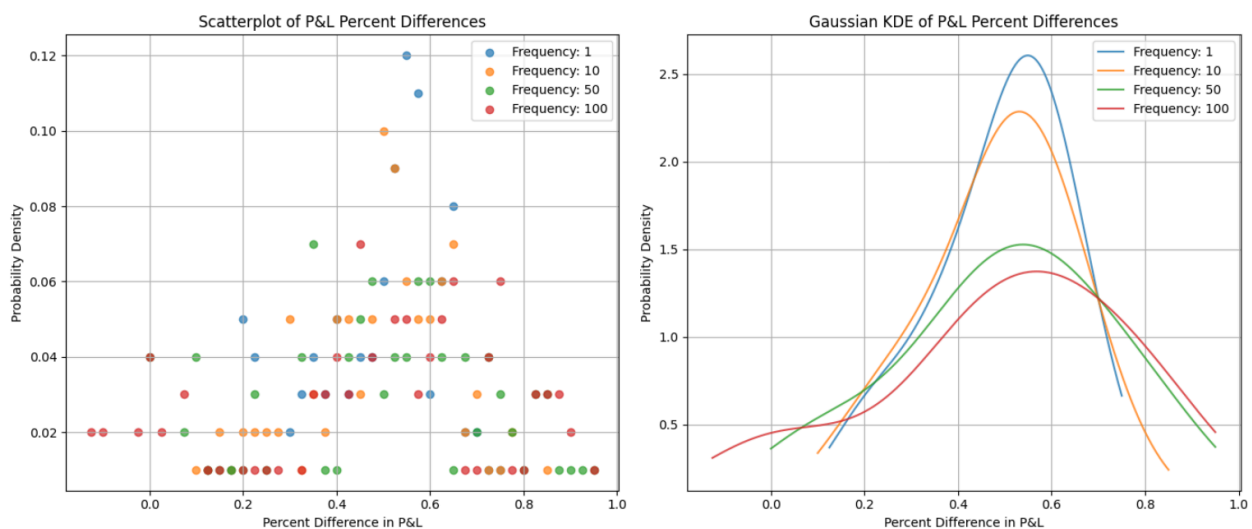


Fig 5: Distribution of percent differences in P&L comparing initial and final portfolio value for constant volatility $\sigma = 0.5$

After investigating the constant volatility scenario, we naturally moved to using GARCH(1,1) to model stochastic volatility. The difference between distributions is even further emphasized in this scenario, where a hedging frequency of 1 shows significantly more stability than a hedging frequency of 100. Furthermore, the distributions are all centered around 0.0, which makes sense given that we expect the value of the portfolio to remain constant over time. We see that, even just visually, our Monte Carlo simulation agrees with theory - delta hedging more frequently decreases gamma exposure. Hence, hedging more frequently assumes less risk.

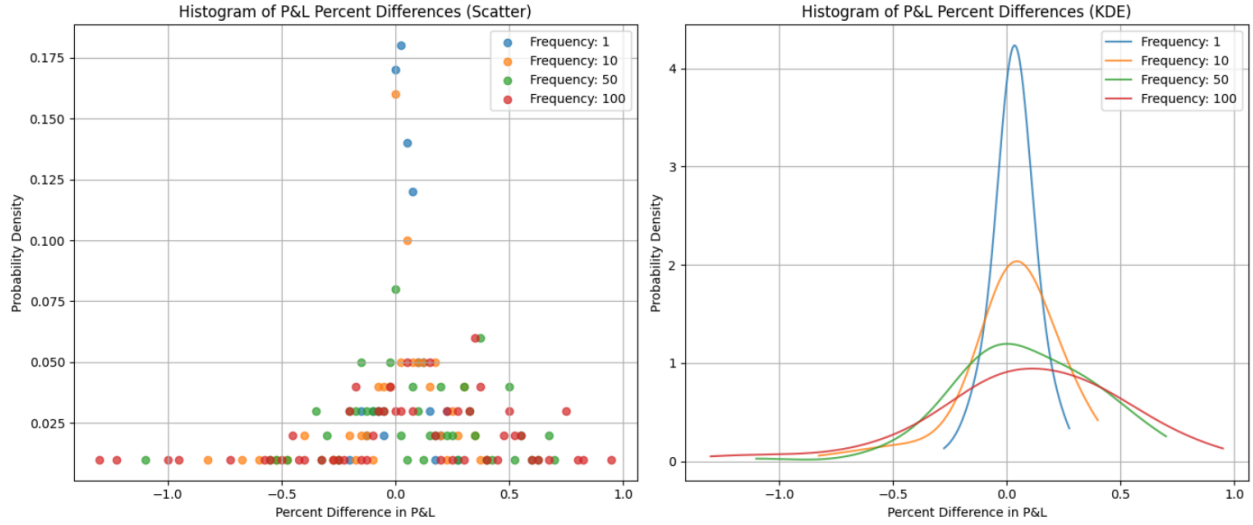


Fig 5: Distribution of percent differences in P&L comparing initial and final portfolio value for $GARCH(1,1)$

Furthermore, we compute risk-adjusted metrics for each of our simulation runs. Based on our hypothesis, we expect both the Sharpe ratio to remain constant and the hedge MSE to increase for increasing hedge frequencies. Table 1 shows the values for the Sharpe ratio, while Table 2 presents the values for the hedge MSE under various volatility regimes.

Volatility Model	Hedging Frequency				
		1	10	50	100
	$\sigma = 0.1$	-0.148	-0.148	-0.147	-0.147
	$\sigma = 0.5$	-0.049	-0.050	-0.048	-0.049
	$GARCH(1,1)$	0.010	0.012	0.021	0.024

Table 1: **Sharpe Ratio** for various volatility models and hedging frequencies

Volatility Model	Hedging Frequency				
		1	10	50	100
	$\sigma = 0.1$	0.164	0.166	0.173	0.177
	$\sigma = 0.5$	6.003	5.990	5.805	5.778
	<i>GARCH(1,1)</i>	63.873	63.311	60.161	57.722

Table 2: Hedge MSE for various volatility models and hedging frequencies

Looking at Table 1, we see that the Sharpe ratio seems to remain consistent across different volatility regimes, although it increases when using the GARCH(1,1) model. This is counter to our expectation that lower frequencies will yield a greater ratio. Instead, a relatively constant Sharpe ratio across different hedging frequencies indicates that while the portfolio's volatility changes, it does so in proportion to its returns. This suggests that hedging frequency does not significantly improve risk-adjusted returns in our specific setup.

Further, looking at Table 2, we observe that hedging MSE behaves differently across volatility regimes. In the low-volatility case, MSE remains relatively small and increases slightly as hedging frequency decreases, suggesting that more frequent hedging marginally improves accuracy. However, in the high-volatility case $\sigma = 0.5$, MSE is significantly larger, yet decreases with less frequent hedging, indicating that excessive rebalancing in volatile environments may introduce inefficiencies. Under the GARCH(1,1) model, MSE is an order of magnitude higher, reflecting the challenges of hedging in dynamic volatility conditions. The downward trend in MSE for lower hedging frequencies in high-volatility settings suggests that hedging too frequently can amplify errors, reinforcing the importance of adjusting hedging strategies based on volatility regimes.

We now ground our analysis in historical data. To truly profit from delta hedging, beyond just staying risk neutral, we need there to be a difference between implied and actual volatility, and we can now work with a real historical simulation of the Russell 2000 prices to recognize how this plays into our results so far. One thing to note is that the actual volatility depends on the window chosen, which can influence some of the interpretations of the results, but they're relevant nonetheless. Towards the end of the implied volatility curve, we can see a huge spike in volatility, which manifests in the overcorrection in all but the every-time-step hedging strategy, further clarifying some of our findings. Moreover, we can see that if we incorporate transaction costs (\$0.01 per share), the frequent hedging strategies perform exponentially worse, owing to the rise in costs, which are determined by every change in delta.

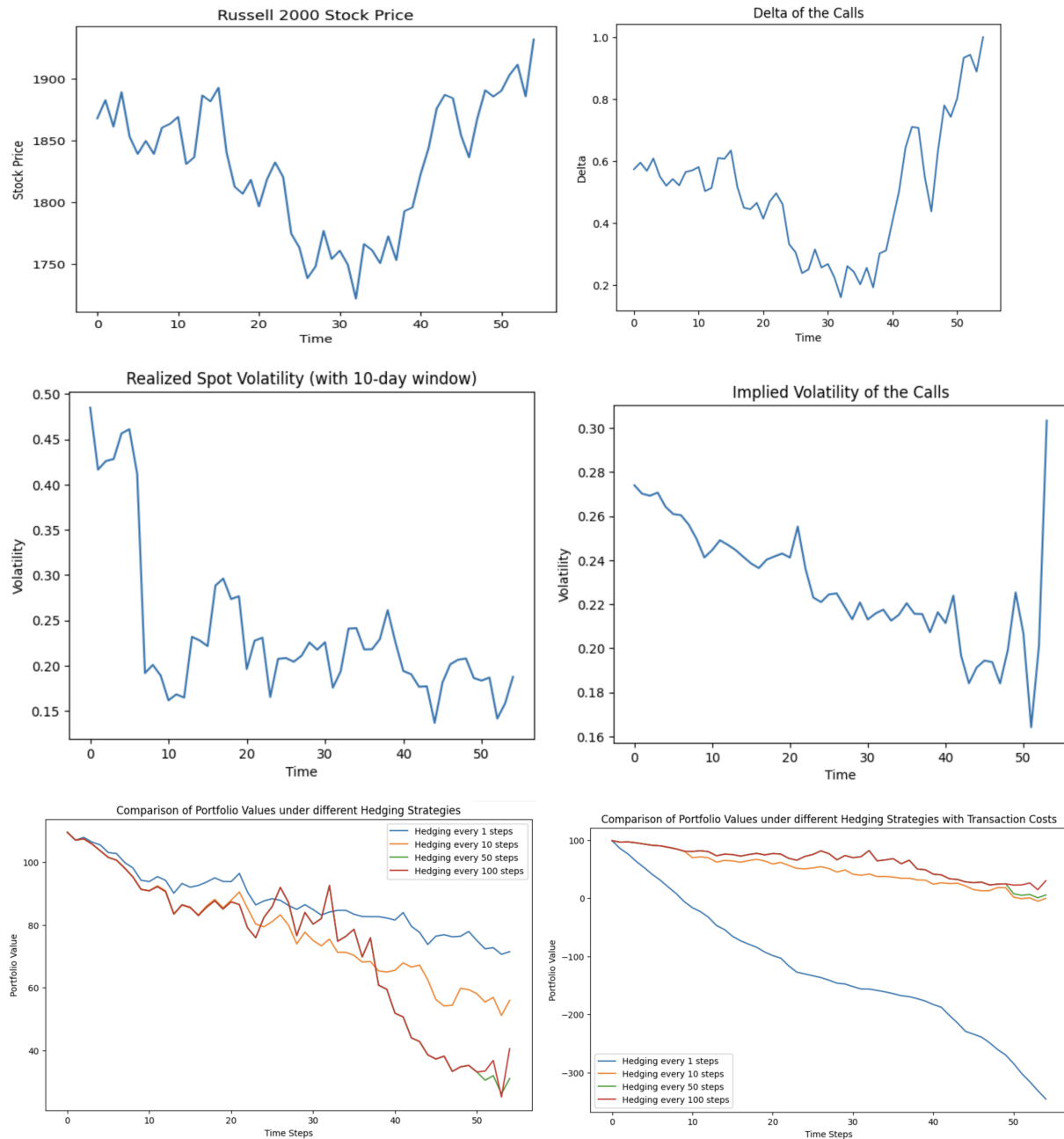


Fig 4: Historical Simulation with RUSSELL 2000

FUTURE WORK

An extension of this project could look at:

- More volatile regimes may be defined by individual stocks rather than indexes, which might require corrections for stock-splits or dividends, and understanding how frequency in these settings might matter by acting before and after the ex-dividend date since you

would short different amounts of shares in both situations. Moreover, different volatility modeling ideas like EGARCH could also be interesting.

- A more realistic transaction setting that doesn't assume a general spread loss and can incorporate something like an order book so that we sell at the best bid and buy at the best ask, making the results more realistic (although converging to something similar to our results)
- Non-liquid calls where the implied volatility might be hugely different from the actual volatility, or non-liquid shares, where infrequent hedging could have a more stable convergence than frequent hedging because of the huge spreads.

CONCLUSION

In conclusion, our findings reinforce the complexity of delta hedging in practical settings and highlight the influence of volatility regimes on hedging effectiveness. Through our simulations, we demonstrated that while continuous hedging remains theoretically optimal under Black-Scholes assumptions, real-world constraints like transaction costs and stochastic volatility make adaptive hedging more viable.

Moreover, hedging frequency plays a crucial role in determining both the risk-adjusted performance and stability of the portfolio. Frequent hedging provides smoother portfolio trajectories, reducing exposure to large fluctuations, but at the cost of higher transaction costs or loss in gamma exposure, which can reduce the profitability of these strategies. By constructing probability distributions, we further find that while frequent hedging reduces variance, it doesn't always optimize risk-adjusted returns when accounting for costs. The tradeoff between hedge accuracy and transaction costs suggests that there is no universally optimal hedging strategy. Instead, an optimal balance exists for different market conditions.

Ultimately, this study underscores the limitations of static hedging strategies and urges deeper research into adaptive strategies, such as regime-switching GARCH or reinforcement learning-based strategies, which can better capture nuances in the market conditions and react accordingly. By acknowledging the interplay between market volatility, transaction costs, and hedge efficiency, practitioners can develop more robust delta hedging strategies tailored to the realities of financial markets.

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Appendix A: PDF Metrics

1. $\sigma = 0.1$

	Mode	Mean	Std. Dev.	Skewness	Kurtosis
1	0.915	0.807	0.110	-0.233	-0.946
10	0.916	0.811	0.118	-0.075	-1.112
50	0.919	0.797	0.114	-0.184	-0.941
100	0.922	0.795	0.137	-0.274	-0.963

2. $\sigma = 0.5$

	Mode	Mean	Std. Dev.	Skewness	Kurtosis
1	0.594	0.456	0.181	-0.209	-1.023
10	0.531	0.452	0.212	0.052	-1.126
50	0.539	0.511	0.272	-0.139	-1.133
100	0.568	0.451	0.306	-0.209	-1.071

3. GARCH(1,1)

	Mode	Mean	Std. Dev.	Skewness	Kurtosis
1	0.035	0.021	0.148	-0.158	-0.864
10	0.044	-0.058	0.314	-0.660	-0.296
50	0.003	0.079	0.380	-0.620	0.557
100	0.112	0.017	0.522	-0.524	-0.106